ZEAL COLLEGE OF ENGINEERING & RESEARCH,

NARHE, PUNE

Department of Computer Engineering

Department of Computer Engineering

COIN201: Internship

ACADEMIC YEAR 2025-26

"AI RESUME ANALYZER"

Presented By:

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1	Darshan Kailas Matre	CO1630
2	Saayali Rajaram Shinde	CO1649
3	Snehal shekhar jagtap	CO1619
4	Shruti Sameer Khadse	CO1653
5	Snehal Ramrao Raut	CO1643
6	Sanchit Suresh Patil	CO1645

Guided By:Prof. A.S.Khaple

Department of Computer Engineering

ZEAL COLLEGE OF ENGINEERING & RESEARCH, NARHE
CERTIFICATE

This is to certify that the Internship Entitled

“AI RESUME ANALYZER”

Submitted by

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Prof. Aparna V. Mote
H.O.D
ComputerEngineering

Dr. Ajit M. Kate
Principal
ZCOER

Internship Supervisor

Acknowledgment

It is our pleasure to acknowledge a sense of gratitude to all those who helped us in the completion of the Internship. I am highly indebted to **Prof.A.S Khaple (Internal Supervisor)** from **Zeal College, Narhe** for their guidance and constant supervision as well as for providing necessary information regarding the internship & also for their support.

I would like to express my gratitude towards **Prof. Aparna V. Mote (HOD Computer Engineering)** for their kind cooperation and encouragement which helped me for providing the required facilities.

Finally, we wish to thank and appreciate all our teachers and friends for their constructive comments, suggestions, and guidance and all those who directly or indirectly helped us in completing this internship.

ABSTRACT

The **AI Resume Analyzer** is a web-based application designed to help job seekers improve the quality of their resumes using Artificial Intelligence. The system allows users to upload their resumes in PDF or DOCX format, after which the application analyzes the content based on important parameters such as skills, education, work experience, keywords, formatting, and overall resume structure.

The application compares the uploaded resume with industry standards and job requirements to identify strengths and areas that need improvement. It then generates a resume score along with detailed feedback and personalized suggestions to enhance the resume's effectiveness. The system also helps users optimize their resumes for Applicant Tracking Systems (ATS), increasing their chances of being shortlisted by recruiters.

The project is developed using **HTML, CSS, JavaScript, PHP, and MySQL**, with AI and Natural Language Processing (NLP) techniques used for resume analysis. The user-friendly interface ensures that users can easily upload resumes, view analysis reports, and access recommendations for improvement.

The AI Resume Analyzer aims to reduce the time and effort required for manual resume evaluation while providing accurate, consistent, and intelligent feedback. This project serves as a valuable tool for students, fresh graduates, and professionals seeking better employment opportunities by helping them create ATS-friendly, high-quality resumes.

Internship Place Details

Activities/Scope	Only for School and College ERP (Enterprise Resource Planning) Software and Technical Workshops forEntrepreneurship Development
Objective of Study	• Learning web development(HTML,CSS) • Improving Python programming skills • Gaining experience with database management • Understanding software development life cycle • Building a portfolio • Building A Responsive Website • Gaining Industry level Knowledge
Supervisor Details (Name, Designation, Company Name, Email-id, Contact number)	

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Introduction

The rapid advancement of Artificial Intelligence (AI) has transformed various industries by automating complex tasks and improving decision-making processes. One of the emerging applications of AI is in the recruitment and hiring process, where intelligent systems assist recruiters in screening and evaluating candidates efficiently. A resume serves as the first impression of a candidate and plays a vital role in determining whether they are shortlisted for an interview. However, many job seekers struggle to create resumes that effectively showcase their qualifications, skills, and experience while meeting the standards of modern Applicant Tracking Systems (ATS). As a result, many deserving candidates fail to secure interview opportunities despite having the required qualifications.

The **AI Resume Analyzer** is a web-based application developed to address this challenge by providing an automated and intelligent resume evaluation system. The application enables users to upload their resumes in PDF or DOCX format and analyzes them using Artificial Intelligence and Natural Language Processing (NLP) techniques. The system evaluates various aspects of the resume, including formatting, skills, educational qualifications, work experience, keyword relevance, resume structure, and ATS compatibility. Based on the analysis, it generates a resume score along with detailed feedback and practical suggestions to help users improve the overall quality of their resumes.

This project has been developed using modern web technologies such as **HTML, CSS, JavaScript, PHP, and MySQL**, providing an interactive, responsive, and user-friendly interface. The front-end ensures a smooth user experience, while the back-end securely manages user information, resume uploads, and analysis reports. The system follows the Agile Software Development Methodology, allowing continuous development, testing, and improvement throughout the project lifecycle.

The AI Resume Analyzer is designed to benefit engineering students, fresh graduates, working professionals, career guidance centers, educational institutions, and recruiters by simplifying the resume evaluation process. It reduces the time and effort required for manual resume screening while delivering accurate, consistent, and personalized feedback. By identifying missing skills, improper formatting, weak keyword usage, and other deficiencies, the application helps users create professional and ATS-friendly resumes that increase their chances of securing interviews and employment.

The internship provided valuable practical experience in software development, database management, web application design, testing, debugging, and project documentation. It also strengthened technical knowledge, analytical thinking, problem-solving skills, and teamwork while demonstrating the practical application of Artificial Intelligence in solving real-world recruitment challenges.

Title/Problem statement/Objective

Title

AI Resume Analyzer

Problem Statement

Many job applicants prepare resumes without understanding industry standards or Applicant Tracking System (ATS) requirements. As a result, qualified candidates often fail to receive interview opportunities because their resumes lack proper formatting, relevant keywords, or adequate presentation of skills and experience. Manual resume evaluation is time-consuming, inconsistent, and difficult to perform for a large number of applicants. Therefore, an intelligent system is required to automatically analyze resumes and provide constructive feedback.

Objective

The main objectives of the project are:

- To develop an AI-based web application for resume analysis.
- To allow users to upload resumes securely.
- To evaluate resumes based on skills, education, experience, formatting, and keywords.
- To generate an overall resume score.
- To provide personalized recommendations for improvement.
- To improve ATS compatibility of resumes.
- To reduce manual effort in resume screening.
- To provide a simple, attractive, and user-friendly interface.
- To maintain resume data securely in the database.
- To help students and professionals prepare job-ready resumes.

Motivation/Scope & Rationale of the Study

Motivation

The increasing competition in the job market has made resume quality one of the most important factors in securing employment. Recruiters receive hundreds of resumes for a single job opening, making manual screening inefficient. Many deserving candidates are rejected simply because their resumes do not meet ATS requirements or fail to highlight relevant skills effectively.

This project was motivated by the need to simplify resume evaluation using Artificial Intelligence. By providing instant analysis and suggestions, the system helps users identify weaknesses and improve their resumes before applying for jobs.

Scope

The project can be used by:

- Engineering students
- Fresh graduates
- Experienced professionals
- Placement cells
- Career guidance centers
- Educational institutions
- Recruitment agencies

Future enhancements may include:

- AI-powered career recommendations
- Job matching
- Interview preparation
- Cover letter generation
- Multi-language resume analysis
- Cloud deployment
- Mobile application support

Rationale of the Study

The rationale behind this study is to explore how Artificial Intelligence and Natural Language Processing can automate resume evaluation and improve recruitment efficiency. The project demonstrates how intelligent software can reduce human effort, increase accuracy, and provide consistent feedback while enhancing employability.

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Daily Activity Report

Day 1

On the first day of the internship, an orientation session was conducted to introduce the internship objectives, project guidelines, and expected outcomes. Various project topics related to Artificial Intelligence, Web Development, and Software Engineering were discussed with the project guide. Existing technologies and current industry trends were explored to identify a project that would be practical, innovative, and relevant to the field of computer engineering. The requirements, feasibility, and expected scope of different project ideas were analyzed, but no final topic was selected on this day.

Day 2

After discussing different ideas with the project guide, the project titled **"AI Resume Analyzer"** was finalized. The project objectives, problem statement, target users, and expected functionalities were thoroughly understood. The overall project scope was defined, including modules such as user registration, login, resume upload, resume analysis, dashboard, job description comparison, and report generation. A development plan was prepared by dividing the project into manageable tasks to be completed within the internship duration.

Day 3

The third day was dedicated to understanding how modern resume screening systems work. Research was conducted on Applicant Tracking Systems (ATS), resume formatting standards, keyword matching techniques, and AI-based resume analysis. Functional and non-functional requirements were identified and documented. References from technical articles, documentation, and research papers were studied to understand how Artificial Intelligence could improve resume evaluation and provide meaningful suggestions to users.

Day 4
The overall architecture of the AI Resume Analyzer was designed. Flowcharts, module diagrams, and the database structure were prepared to understand the workflow of the application. Required software such as XAMPP, Visual Studio Code, PHP, and MySQL was installed and configured. The project folder was created, the database was initialized, and tables were designed for storing user information, uploaded resumes, and analysis reports.

Day 5
The development of the website began with designing the homepage. HTML and CSS were used to create a professional, responsive, and user-friendly interface. The navigation bar, hero section, footer, and other layout components were developed. The website design was made responsive to ensure proper viewing on different screen sizes. Attention was given to creating a clean and attractive user interface that would provide a positive user experience.

Day 6
The Registration and Login modules were developed using PHP and MySQL. Database connectivity was established to securely store user information. Form validation was implemented to ensure that users entered valid details during registration and login. Session management was introduced to provide secure access to authenticated users. Basic testing was performed to verify that user authentication worked correctly.

Day 7
A Resume Upload feature was developed to allow users to upload resumes in PDF and DOCX formats. File validation techniques were implemented to accept only supported file types and prevent invalid uploads. Uploaded resumes were securely stored, and proper file management techniques were used. The upload functionality was tested using different resume files to ensure reliability and accuracy.

Day 8
The core functionality of the project was developed on this day. The Resume Analysis module was implemented to evaluate uploaded resumes based on predefined parameters such as skills, education, work experience, formatting, and relevant keywords. A resume scoring mechanism was designed to generate an overall score along with improvement suggestions. This module demonstrated the practical use of Artificial Intelligence concepts in simplifying the resume evaluation process.

Day 9
Additional pages such as Dashboard, Job Description, About, and Contact were developed and integrated with the main application. The dashboard was designed to display user information, uploaded resumes, and analysis reports. Navigation between all pages was verified to ensure smooth operation. Overall website consistency, layout.

Day 10

All developed modules were integrated into a single application. Front-end pages were connected with the PHP back-end and MySQL database to ensure proper communication between different components. Database operations such as inserting, retrieving, and updating records were tested. The overall workflow of the application was verified from user registration to resume analysis.

Day 11

The appearance of the website was improved by refining fonts, colors, spacing, buttons, icons, and responsive layouts. Minor errors affecting navigation, alignment, and user interaction were identified and corrected. The overall usability of the website was enhanced to provide a professional look and improve the user experience.

Day 12

Comprehensive testing was carried out on all modules to ensure that the application functioned correctly. Functional testing, integration testing, and user interface testing were performed. Issues related to database connectivity, login, resume upload, and analysis were resolved. Performance optimization techniques were applied to improve the speed and stability of the application.

Day 13

The complete AI Resume Analyzer application was reviewed and finalized. All modules were tested again to ensure they functioned without errors. Implementation screenshots were captured, project documentation was updated, and all project files were organized. Necessary improvements suggested during testing were incorporated before the final submission.

Day 14

The internship report was completed by documenting the project objectives, methodology, implementation details, results, and conclusion. A PowerPoint presentation was prepared containing the project introduction, system architecture, methodology, implementation snapshots, testing results, and future scope. The complete project was reviewed, and presentation delivery was practiced to ensure confidence during the final evaluation.

Day 15

On the final day of the internship, the **AI Resume Analyzer** project was presented before the faculty members. The project objectives, problem statement, system architecture, implementation process, technologies used, application features, testing results, and future enhancements were explained in detail. A live demonstration of the application was given, showing user registration, login, resume upload, analysis, and report generation. Questions asked by the faculty were answered confidently, and valuable feedback was received. The internship concluded successfully with the submission of the project report and presentation.

Methodological Details

The project followed the **Agile Software Development Methodology**, where the work was completed in small phases with continuous testing and improvement.
The development methodology included:

Requirement Analysis

Understanding user needs, identifying system requirements, and defining project objectives.

System Design

Preparing database design, user interface layouts, navigation structure, and overall system architecture.

Database Design
Creating tables for:

- Users
- Resume Uploads
- Job Descriptions
- Analysis Reports
- Contact Messages
- Admin

Development

Developing various modules including:

- Registration
- Login
- Resume Upload
- Resume Analysis
- Dashboard
- About Page

Testing

Performing:

- Unit Testing
- Integration Testing
- Functional Testing
- User Interface Testing
- Performance Testing

Deployment

Running the application using XAMPP localhost environment and preparing it for future deployment.

Results / Analysis / Inferences and Conclusion

Results

The developed AI Resume Analyzer successfully performs the following tasks:

- User registration and login
- Resume upload
- Resume evaluation
- Resume scoring
- Feedback generation
- Dashboard management
- Secure database storage

Analysis

The system analyzes resumes based on:

- Skills
- Educational qualifications
- Experience
- Resume formatting
- Keyword matching
- ATS compatibility
- Resume completeness

The generated feedback enables users to improve resume quality effectively.

Inferences

The project demonstrates that AI can significantly improve resume evaluation by providing quick, consistent, and objective analysis. Automation reduces recruiter workload while helping candidates prepare stronger resumes.

Conclusion

The AI Resume Analyzer successfully fulfills its objectives by offering intelligent resume assessment through a user-friendly web application. The project integrates web technologies with AI concepts to simplify resume evaluation and provide valuable recommendations. It highlights the practical application of Artificial Intelligence in recruitment and career development.

Suggestions

The following improvements can be implemented in future versions:

- Integrate advanced NLP models.
- Add ChatGPT-based resume suggestions.
- Include job recommendation features.
- Support multiple languages.
- Enable cloud storage.
- Provide recruiter dashboard.
- Add interview preparation module.
- Generate cover letters automatically.
- Integrate LinkedIn profile analysis.
- Develop Android and iOS applications.

Attendance Record

The attendance record contains the daily presence of the intern throughout the internship period. It includes:

- Date
- Day
- In-Time
- Out-Time
- Total Working Hours

- Attendance Status
- Supervisor Signature

Maintaining an attendance record demonstrates regular participation, punctuality, discipline, and commitment during the internship.

Acknowledgement

I express my sincere gratitude to my internship organization for providing me with the opportunity to work on the **AI Resume Analyzer** project. I am thankful to my project guide, mentors, and faculty members for their valuable guidance, continuous encouragement, and constructive suggestions throughout the internship.

I also extend my heartfelt thanks to my college, classmates, friends, and family members for their constant support and motivation. Their encouragement helped me successfully complete this internship and gain valuable technical and professional experience.

List of References

The project was developed with the help of the following resources:

- HTML5 Documentation
- CSS3 Documentation
- JavaScript Documentation
- PHP Documentation
- MySQL Documentation
- XAMPP Documentation
- Git and GitHub Documentation
- Artificial Intelligence and NLP research papers
- Resume writing guidelines
- Applicant Tracking System (ATS) resources
- IEEE research papers
- Technical blogs and online tutorials

Overall Result

The **AI Resume Analyzer** project was successfully developed and achieved all the objectives defined at the beginning of the internship. The application provides a simple, secure, and user-friendly platform where users can register, log in, upload their resumes, and receive an analysis based on important resume parameters. The system was developed using **HTML, CSS, JavaScript, PHP, and MySQL**, with **XAMPP** serving as the local development environment.

All the major modules, including User Registration, Login, Resume Upload, Resume Analysis, Dashboard, Job Description, About, and Contact pages, were implemented and integrated successfully. The application was tested thoroughly, and all identified issues related to database connectivity, file uploads, navigation, and user authentication were resolved.

The resume analysis module evaluates resumes based on factors such as skills, educational qualifications, work experience, formatting, and keyword relevance. It generates a resume score and provides suggestions that help users improve their resumes and increase their chances of meeting Applicant Tracking System (ATS) requirements.

The project demonstrated that an AI-assisted resume analysis system can reduce manual effort, provide consistent feedback, and help job seekers prepare professional resumes. The developed application performed as expected and met the functional requirements identified during the planning stage. Overall, the project was completed successfully within the given timeline and produced the expected outcomes.

Conclusion

The **AI Resume Analyzer** is an innovative web-based application that simplifies the process of resume evaluation by combining web technologies with Artificial Intelligence concepts. The project successfully provides users with an efficient platform to upload resumes, analyze their content, and receive valuable feedback for improvement.

During the development of this project, practical knowledge of software development, database management, web application design, testing, debugging, and documentation was gained. The project also strengthened skills in problem-solving, logical thinking, and implementing complete software solutions.

The developed system fulfills its intended objectives by providing accurate resume evaluation, improving ATS compatibility, and assisting students, fresh graduates, and professionals in creating stronger resumes. It offers a user-friendly interface and reliable performance, making it a useful tool for career development.

Although the current version successfully meets the project's objectives, there is scope for future enhancements such as integrating advanced Natural Language Processing (NLP), Machine Learning algorithms, job recommendation features, interview preparation modules, cloud deployment, and multilingual support. These enhancements can further improve the application's intelligence, accuracy, and usability.

In conclusion, the AI Resume Analyzer successfully demonstrates the practical application of Artificial Intelligence in the recruitment domain. It serves as an effective solution for improving resume quality while providing valuable learning experience in designing and developing a real-world software application.